

A Novel Aux/IAA28 Signaling Cascade Activates GATA23-Dependent Specification of Lateral Root Founder Cell Identity

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Summary

Background: Lateral roots are formed at regular intervals along the main root by recurrent specification of founder cells. To date, the mechanism by which branching of the root system is controlled and founder cells become specified remains unknown.

Results: Our study reports the identification of the auxin regulatory components and their target gene, *GATA23*, which control lateral root founder cell specification. Initially, a meta-analysis of lateral root-related transcriptomic data identified the *GATA23* transcription factor. *GATA23* is expressed specifically in xylem pole pericycle cells before the first asymmetric division and is correlated with oscillating auxin signaling maxima in the basal meristem. Also, functional studies revealed that *GATA23* controls lateral root founder cell identity. Finally, we show that an Aux/IAA28-dependent auxin signaling mechanism in the basal meristem controls *GATA23* expression.

Conclusions: We have identified the first molecular components that control lateral root founder cell identity in the *Arabidopsis* root. These include an IAA28-dependent auxin signaling module in the basal meristem region that regulates *GATA23* expression and thereby lateral root founder cell specification and root branching patterns.

Introduction

Higher plants initiate the majority of their adult organs following embryogenesis. *Arabidopsis* lateral root development is used as a model to study the signals and genes that regulate postembryonic organogenesis in plants. In *Arabidopsis*, lateral roots originate exclusively from pericycle cells at the xylem pole. Lateral root initiation can be triggered by the plant hormone auxin and auxin-like molecules [1, 2]. Mechanical stimuli have also recently been shown to be capable of inducing lateral root primordia [3–5]. The actual contribution of these mechanical stimuli to the endogenous mechanisms remains unclear, but in all cases, a localized auxin accumulation pattern could be correlated with the activation of lateral root initiation [6].

The molecular mechanism for auxin-regulated lateral root initiation is well studied and is dependent on an auxin-triggered degradation of the labile repressor SOLITARY ROOT (SLR/IAA14) and subsequent derepression of AUXIN RESPONSE FACTORS ARF7 and ARF19, activating downstream gene expression [7, 8]. Consequently, mutating these key players dramatically affects lateral root initiation [7, 8]. Recently, a second auxin signaling module, based on the action of BODENLOS (BDL/IAA12) and MONOPTEROS (MP/ARF5), has been shown to act after the SLR/ARF7/ARF19 module during lateral root initiation [9]. Also, several additional regulatory genes in lateral root initiation and subsequent lateral root emergence processes have been identified [2, 10]. Despite these recent advances, little is known about the molecular mechanisms determining how pairs of xylem pole pericycle (XPP) cells are selected to become lateral root founder cells and undergo cell division.

De Smet et al. recently suggested that XPP cells could adopt founder cell identity in a zone neighboring the meristem, designated as the basal meristem [11]. Starting 10 hr after germination, oscillating peaks of *pDR5::GUS* expression could be observed in this zone with an interval of ~15 hr [11]. These recurrent peaks of auxin responsiveness in protoxylem cells of the basal meristem were interpreted to be required to specify neighboring XPP cells as lateral root founder cells. Although this recurrent auxin signaling event appears to specify sets of XPP cells as founder cells, its molecular basis has not been determined thus far. Here, we report the identification of the *GATA23* transcription factor, which plays a key role in specifying pericycle cells to become lateral root founder cells prior to lateral root initiation. Our study reveals that *GATA23* expression correlates with the auxin response oscillations in the basal meristem and is controlled by the auxin regulatory gene *IAA28*.

Results

Identification of the Novel Putative Lateral Root Regulatory Gene *GATA23*

To identify regulatory genes involved in the very early steps of lateral root initiation, we performed a meta-analysis of all relevant available transcriptomic data sets (see Figure S1 available online) [8, 12–14] using the VisualRTC [15] spreadsheet

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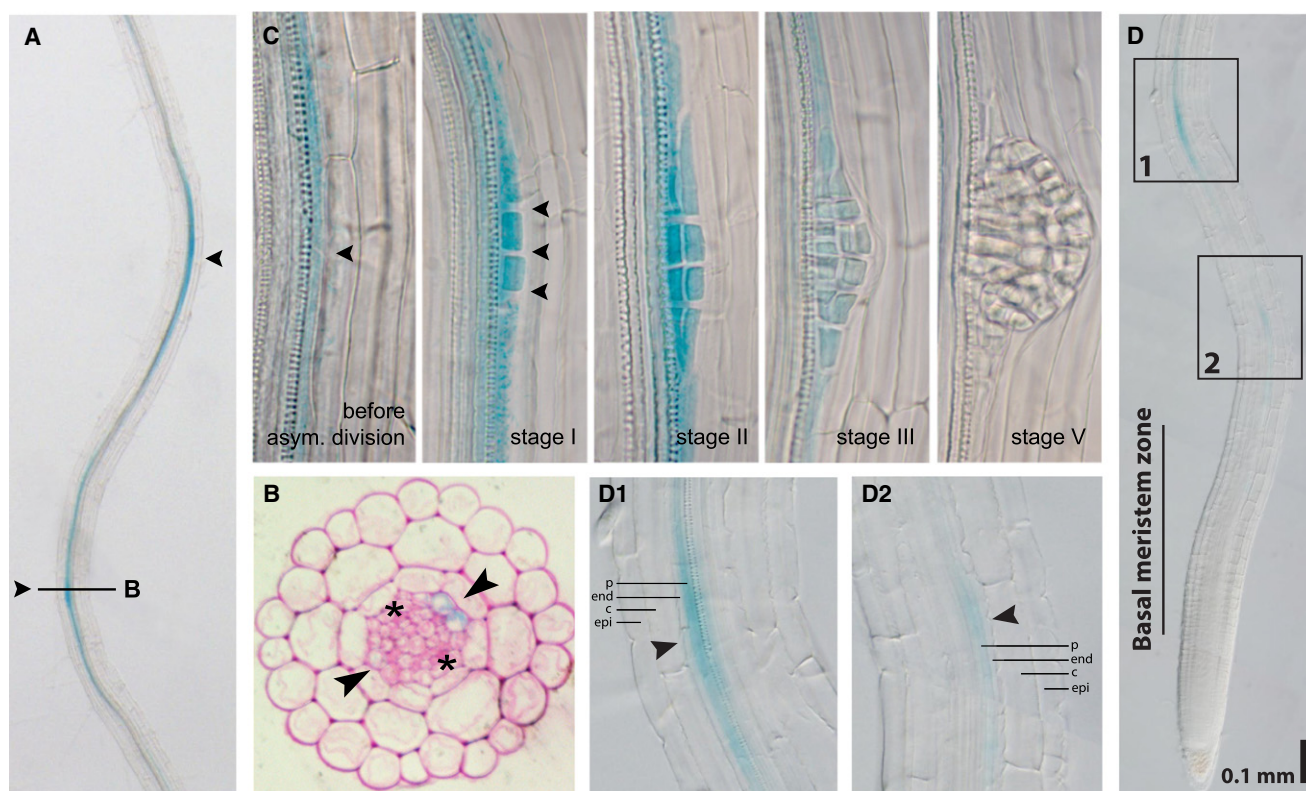


Figure 1. GATA23 Expression Analysis

(A) pGATA23::GUS expression in the mature root at sites of lateral root initiation (arrowheads). The horizontal black line represents the location of the section shown in (B).
(B) Section through a stage I lateral root primordium showing pGATA23::GUS expression at one side of the xylem pole pericycle (XPP). Asterisks and arrowheads indicate phloem pole and XPP cells, respectively.
(C) Detailed expression analysis of GATA23 during stages of lateral root initiation.
(D) Before the first asymmetric division, GATA23 is expressed in patches in the root (arrowheads in D1 and D2). The vertical black line in (D) indicates the basal meristem. (D1) and (D2) show magnifications of boxed areas in (D) (epi, epidermis; c, cortex; endo, endodermis; p, pericycle).

application (see [Supplemental Experimental Procedures Part A](#)). The meta-analysis employed the following selection criteria: differential expression between xylem and phloem pole pericycle cells [14, 16], absence from other radial layers [16], auxin-regulated expression [8, 12], and involvement in asymmetric divisions [13] but not in general cell-cycle phase transitions [17]. Although a nonstringent approach (low p value) guaranteed a maximum of candidate genes, only two genes were recovered, including GATA23 (Figure S1). Because GATA-type transcription factors are known to be critical players in regulatory networks that determine the specification of cell fates during mammalian development [18–20] and because of GATA23's strongest differential expression, we investigated GATA23 spatial and temporal expression during lateral root development in more detail.

GATA23 Is Specifically Expressed in Xylem Pole Pericycle Cells before the First Asymmetric Division

A transgenic line expressing the GUS reporter under the regulation of the GATA23 promoter was created (termed pGATA23::GUS; see [Experimental Procedures](#)), and its expression was found to be root specific (Figure 1A) and limited to pericycle cells at xylem poles (Figures 1B and 1C). pGATA23::GUS activity was absent from the root apical meristem (Figure 1D) and was confined to early stages of lateral root formation: strong expression in stage I and II primordia [21],

reduced staining in stage III, and absent at later stages (Figure 1C). Interestingly, pGATA23::GUS expression was also observed in patches of XPP cells near the root tip in a zone lacking lateral root primordia, close to the basal meristem (Figures 1D–1D2). This spatial pattern of expression suggests that GATA23 is involved in a process occurring prior to lateral root initiation.

To investigate cellular events and auxin activity preceding the first asymmetric division, we constructed a p35S::FH6-GFP × p35S::H2B-RFP × pDR5::NLS-GFP triple reporter line that marks plasma membranes, nuclei, and auxin maxima, respectively (Figure S2A). As reported previously, auxin accumulates in two neighboring XPP cells [22], visualized by pDR5::GUS expression, causing a red-to-yellow color shift in nuclei of the central tissues (Figure 2A). Next, the nuclei of two adjacent pericycle cells situated on a longitudinal cell file were observed to migrate toward the common cell wall (Figure 2A, time 1:55–3:15). The first asymmetric anticlinal divisions occurred within 4–7 hr after the onset of nuclear repositioning and resulted in formation of a single-layered structure of shorter and longer cells (stage I primordia) [21] (Figure 2A; [Movie S1](#); [Figure S2B](#)). Following several additional anticlinal divisions and within approximately 8–13 hr after the onset of nuclear migration, the division plane rotated, giving rise to a second cell layer, thus forming stage II primordia (Figure 2A; [Figure S2B](#)).

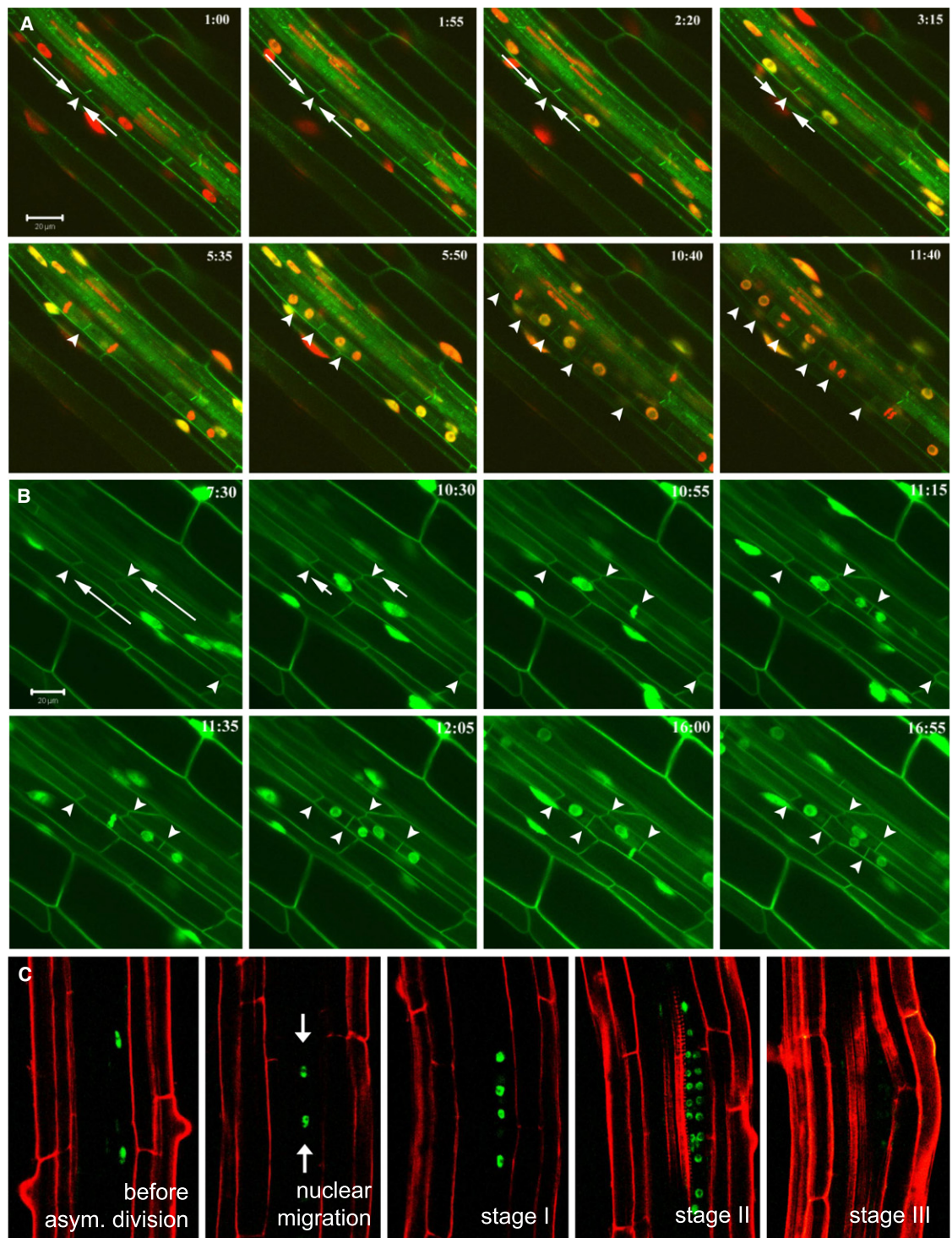


Figure 2. GATA23 Reveals a Nuclear Migration Event

(A) Confocal live imaging of the nuclear migration event in the XPP of a wild-type root using a p35S::FH6-GFP × p35S::H2B-RFP × pDR5::NLS-GFP triple marker line. Arrows indicate movement of the nuclei; arrowheads indicate common cell walls.

(B) Confocal live imaging of the nuclear migration event in the XPP of *slr-1* roots treated with 10 µM naphthalene-1-acetic acid using a p35S::LT16B-GFP × p35S::H2B-YFP line [27]. Time of live imaging is indicated in upper right corner of each frame. Scale bar represents 20 µm.

(C) Confocal images of pGATA23::NLS-GFP counterstained with propidium iodide (red).

Nuclear Migration Is Affected in Auxin Signaling Mutants and Is Preceded by GATA23 Expression

We next investigated whether polar nuclear migration is SLR/IAA14 dependent. Under control conditions, no polar

movement could be observed in the *slr-1* mutant background (data not shown). However, after prolonged auxin treatment, polar movement of nuclei and asymmetric cell divisions did occur, but only in single XPP cells, never simultaneously in

pairs of neighboring cells and never resulting in formative divisions that normally precede the formation of a functional lateral root primordium (Figure 2B; Movie S2; Figure S2B). Similar defects could be observed in other mutant backgrounds that are severely affected in lateral root development (Table S3; Figure S3), such as the *arf7arf19* double mutant. These observations suggest that the coordinated nuclear migration of two neighboring XPP nuclei depends on Aux/IAA and ARF-dependent signaling and is an absolute prerequisite for proper primordium initiation and formation of lateral roots.

In vivo analysis of GATA23 expression employing a nuclear-targeted GFP reporter (termed pGATA23::NLS-GFP) revealed an almost simultaneous migration of two nuclei toward the common cell wall of two neighboring XPP cells in GATA23 expression patches just before the first asymmetric division (Figure 2C; Movie S3). No differences could be observed with a pGATA23::GATA23-GFP translational fusion construct (Figure S2C), arguing for a cell-autonomous action of GATA23. Hence, the GFP reporters reveal that GATA23 is expressed prior to the synchronized migration of nuclei in pairs of XPP cells that precedes lateral root initiation.

GATA23 Controls Lateral Root Founder Cell Identity

To evaluate the function of GATA23 during lateral root founder cell specification, and because available T-DNA insertion lines either showed no reduction in mRNA levels or had insertions in the 3' untranslated region or only 19 bp before the stop codon (data not shown), we chose to construct RNA interference (RNAi) lines based on a gene-specific tag for GATA23. Although no full knockout lines were obtained, expression was reduced to about 30% of normal levels and this reduction was specific for GATA23, but not its homologs (Figures S4A–S4C), supportive for the specificity of the GATA23 RNAi line. Using this line, we observed a strong reduction in the number of stage I and II primordia and an overall decrease in the number of emerged and nonemerged primordia (Figure 3A; Figure S4D). In contrast, overexpression of GATA23 using the XPP-specific J0121 GAL4 driver line resulted in a very significant increase in the amount of non-emerged primordia compared to the J0121>>Col-0 control (Figure 3A; Figure S4D). Detailed morphological analysis revealed that this was due to a strong increase of stage I and II primordia (Figures 3A and 3B). Moreover, in the J0121>>UAS::GATA23 line, primordia could be observed that were in close proximity to each other at considerably higher frequencies compared to control roots (Figure S4E). Sporadically, pairs of primordia were observed that resulted in two emerged primordia next to one other (Figure S4F). These ectopic primordia imply that excessive founder cell specification events took place in the basal meristem of the J0121>>UAS::GATA23 line. Hence, GATA23 appears to promote initiation by controlling founder cell identity of XPP cells.

GATA23 Expression in Founder Cells Is Correlated with Oscillating Auxin Signaling Maxima in the Basal Meristem

Our functional studies are consistent with GATA23 being involved in the specification of lateral root founder cells. To further understand the relationship between GATA23 and auxin-responsive gene expression in the basal meristem, we monitored the temporal expression of pDR5::GUS and pGATA23::GUS reporters. Synchronously germinated

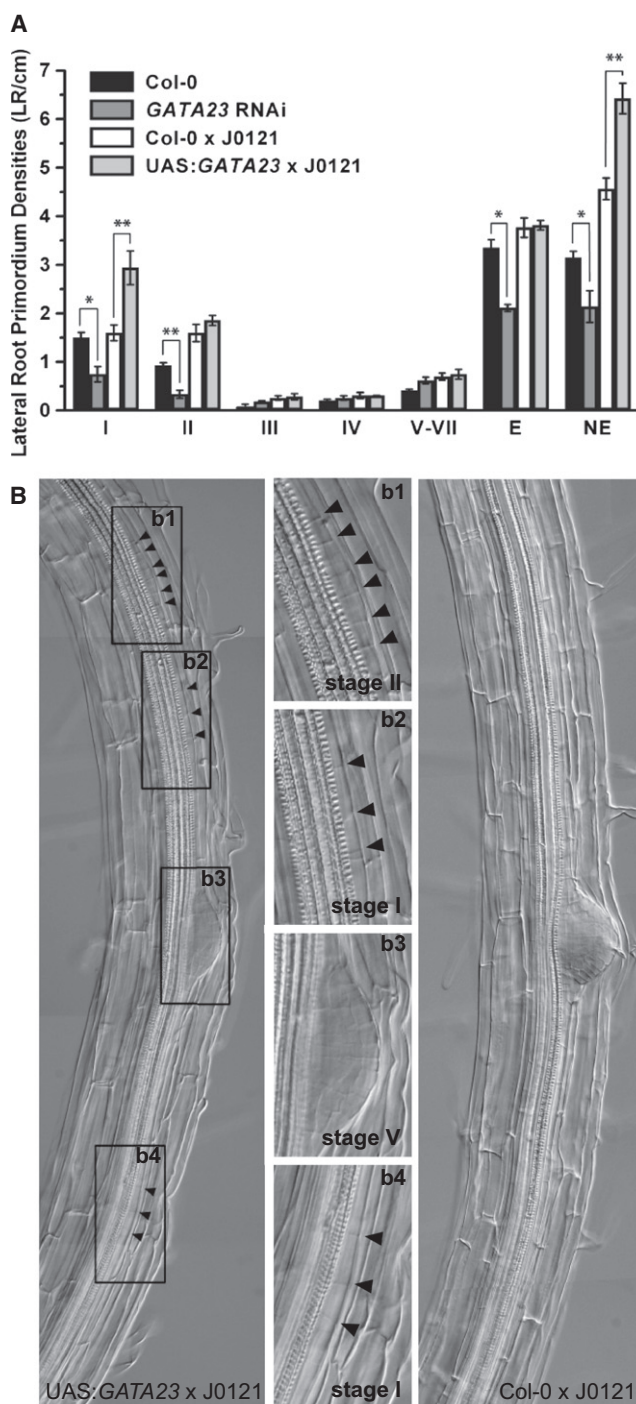


Figure 3. GATA23 Knockdown and Transactivation Phenotypes

(A) Detailed staging of lateral root stage densities (lateral roots/cm root; $n > 10$) of the indicated lines 10 days after germination. E, emerged lateral roots; NE, sum of nonemerged stages. Error bars represent means $\pm$ standard error of the mean (SEM). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ by two-sided t test; $n > 10$.

(B) Example of J0121>>GATA23 root showing ectopic initiation events resulting in excessive lateral root primordia (left and insets in middle) compared to a Col-0 $\times$ J0121 control cross (right).

seedlings [11] were evaluated for pDR5::GUS and pGATA23::GUS activity every 5 hr from germination until 50 hr after germination (HAG) (Figures S5A–S5C). As reported

previously [11], an oscillating *pDR5::GUS* maximum was observed in the protoxylem cells of the basal meristem with an interval of about 15 hr and a first peak at 10 HAG (Figure 4A, blue arrows). *GATA23* promoter activity was observed in specific patches along the root before the first asymmetric division. These patches increased in number over time, reaching a maximum of about three patches per root at 50 HAG (Figure 4A; Figure S5B). The increase in the number of *pGATA23::GUS*-positive patches was not continuous but followed a cumulative stepwise function (see Supplemental Experimental Procedures Part B for statistical analysis), increasing by an average of one patch per root about 10 hr after each *pDR5::GUS* peak (Figure 4A, blue arrows), indicative of a correlation between the auxin response in the basal meristem and *GATA23* expression. We next assessed whether *pDR5::GUS* and concomitant *GATA23* expression in XPP cells before the first asymmetric division were also dependent on auxin signaling. Three-day-old seedlings carrying *pDR5::GUS* and *pGATA23::GUS* constructs were incubated for 24 hr on medium containing 25 μ M PEO-IAA, an anti-auxin that binds the TIR1/AFB auxin receptors [23] and therefore inhibits TIR1/AFB-dependent auxin signaling. Interestingly, newly grown parts of the root no longer expressed *pGATA23::GUS* compared to mock-treated roots (Figure 4B). Furthermore, the oscillating *pDR5::GUS* expression in the basal meristem was also strongly reduced or absent (Figures 4B and 4C). These data not only suggest that *pDR5::GUS* expression in the basal meristem is controlled by TIR1-dependent auxin signaling but also argue that auxin signaling in the basal meristem is causal for *GATA23* expression in XPP founder cells above the basal meristem.

An Aux/IAA28-Dependent Auxin Response Controls *GATA23* Expression in XPP Cells

We have shown that the auxin response in the basal meristem is dependent on TIR1- and AFB-mediated auxin signaling (Figures 4B and 4C). TIR1 regulates auxin responses by targeting the degradation of Aux/IAA repressor proteins. Previously, we demonstrated that the auxin response in the basal meristem is SLR/IAA14 independent [11], which is corroborated by the absence of *SLR/IAA14* expression in this part of the root [12]. To identify which Aux/IAA is involved, we quantified *GATA23* expression in available *aux/iaa* mutants. Relative *GATA23* expression levels were unaltered in most *aux/iaa* mutant backgrounds (Figure 4D; Figure S5D) and were increased in the *bd1/iaa12* and *shy2/iaa3* gain-of-function mutants that develop clusters of lateral root primordia (Figure S5E). The relative expression level (Figure 4D) and promoter activity (Figure 4E) were only reduced in the *iaa28-1* mutant background.

Lateral root development is strongly reduced in the *iaa28-1* gain-of-function mutant background [24] (Figure 4F; Figure S5F), and nuclear migration is also strongly affected (Table S3). *IAA28* is expressed in the zone corresponding to the basal meristem [24] (Figure 5B). In addition, the percentage of *pDR5::GUS* plants showing staining in the basal meristem was strongly reduced in the *iaa28-1* gain-of-function mutant background (Figure 4G). Taken together, these data suggest the Aux/IAA28-dependent auxin signaling module regulates the auxin response in the basal meristem.

To test whether *GATA23* acts downstream of *IAA28* to control lateral root founder cell identity, we initially compared expression of the two genes in longitudinal sections of the root: the meristem (MS), the rapid elongation zone (EZ,

corresponding to the basal meristem region), the late elongation (deceleration) zone (DZ), mature root (MR), and the lateral root zone (LRZ) (see Supplemental Experimental Procedures Part C for details). Importantly, both *GATA23* and *IAA28* expression levels are already increased in the zone corresponding to the basal meristem (EZ) and peak in the zone just above the basal meristem (DZ) (Figures 4H and 4I). A similar trend could be observed (Figure S5G) in the publicly available data set describing expression profiles along the longitudinal axis of the root [16], confirming our results.

Finally, to determine whether *GATA23* activity downstream of *IAA28* is functionally required for lateral root initiation, we ectopically expressed *GATA23* in the *iaa28-1* mutant background. Transactivation of *GATA23* expression in *iaa28-1* XPP cells was able to rescue the mutant's lateral root phenotype (Figure 4J). This result provides compelling evidence that *GATA23* acts in the same pathway as *IAA28* and represents an important factor controlling lateral root founder cell identity.

A Set of Five ARFs Are Involved in the IAA28-Dependent Signaling Module

Aux/IAA repressors such as *IAA28* regulate auxin responses by interacting with inhibiting transcription factors termed auxin response factors (ARFs). Yeast two-hybrid (Y2H) interaction studies revealed that *IAA28* is capable of interacting with five ARF proteins: ARF5, ARF6, ARF7, ARF8, and ARF19 (Figure 5A). Interestingly, promoter-GUS/GFP localizations showed that all five ARFs are expressed to some extent in the basal meristem region (Figure 5B) and thus are potentially important for the novel *IAA28*-dependent auxin signaling module in the basal meristem. Nevertheless, because the auxin inducibility of *GATA23* is almost completely absent in the *arf7arf19* double mutant (Figure S1), ARF7 and ARF19 are likely the most relevant ARFs in controlling this auxin signaling module.

In summary, our data show that a novel auxin signaling module based on *IAA28* and ARF5, 6, 7, 8, and/or 19 acts in the basal meristem. As a consequence, *GATA23* expression starts to be induced above the basal meristem region and is first visible with marker lines as patches in the zone just above the basal meristem region.

Discussion

Although the molecular basis of lateral root initiation has been extensively studied, little is known about the preinitiation events [2]. However, it is during this phase of lateral root formation that the most crucial decisions for root architecture are taking place. This is achieved through the specification of lateral root founder cells. In the absence of known regulators and/or reporter lines for lateral root founder cells, a better understanding of this process has remained out of reach. Here we have identified *GATA23* as a first molecular component of the founder cell specification process. Its patchy expression pattern along the young root axis in subsets of XPP cells reflects an oscillating induction pattern that appears to be correlated with the earlier reported oscillating auxin response maximum in the basal meristem of the root tip [11]. We further were able to pinpoint the involvement of a potential auxin response module controlled by the repressor protein *IAA28* that translates the oscillations of the auxin response into downstream transcriptional networks through *GATA23* (Figure 5C, module 1). The involvement of *IAA28* in lateral

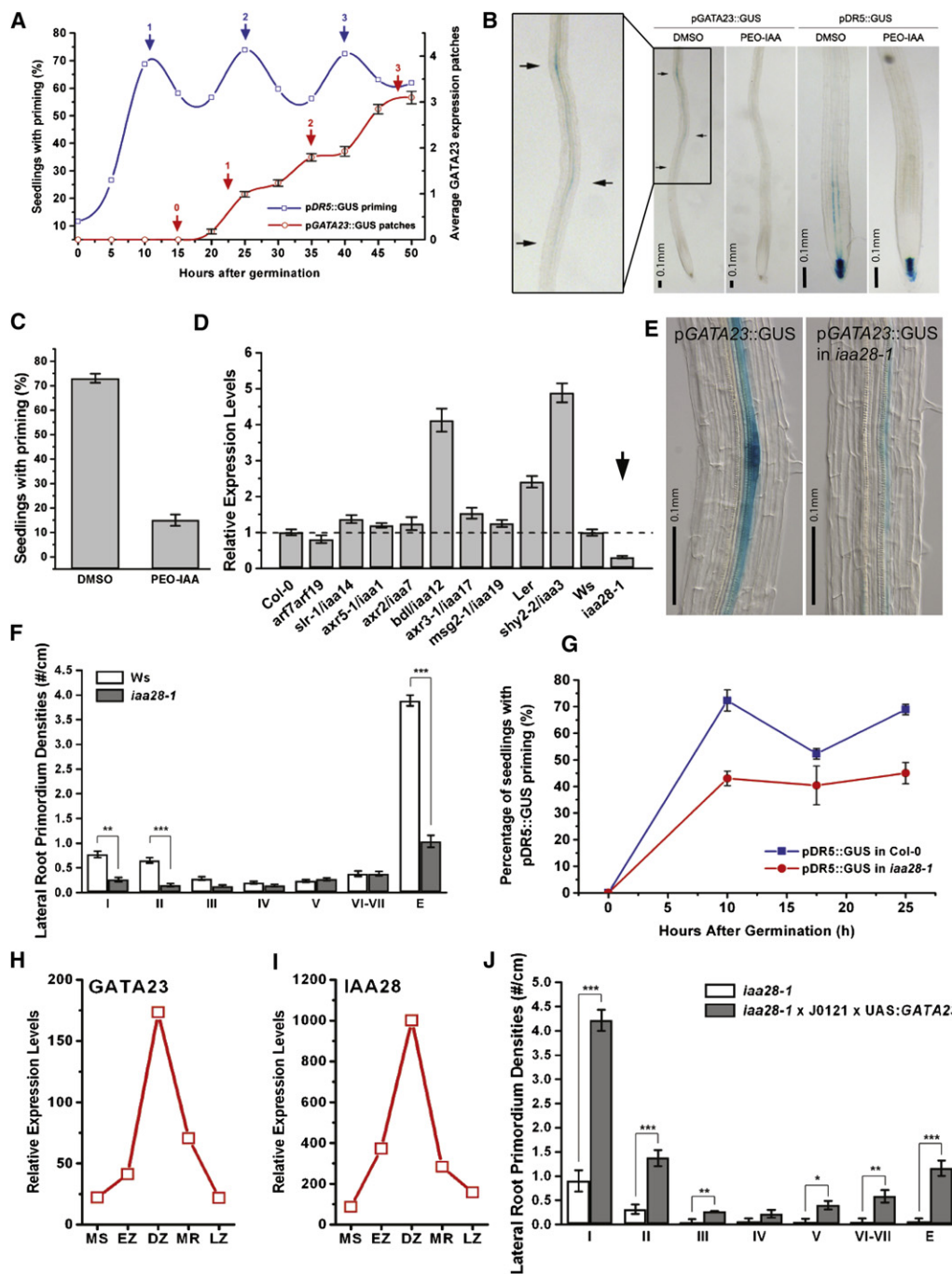


Figure 4. GATA23 Controls Lateral Root Founder Cell Specification and Is IAA28 Dependent

(A) Time-lapse analysis from 0 to 50 hr after germination showing the percentage of seedlings with pDR5::GUS staining (blue) in the basal meristem region and the average cumulative number of patches expressing pGATA23::GUS (red).

(B) pDR5::GUS and pGATA23::GUS expression in roots treated with 25 μ M PEO-IAA (anti-auxin) or mock treated (DMSO) for 24 hr (arrowheads show pGATA23::GUS expression patches).

(C) Percentage of seedlings ($n \geq 100$) with pDR5::GUS expression in the basal meristem at 25 hr after germination when treated with 25 μ M PEO-IAA or mock treated (DMSO).

(D) Relative GATA23 expression levels in auxin signaling mutants compared to their respective wild-type.

(E) pGATA23::GUS expression in wild-type and *iaa28-1* mutant background.

(F) Lateral root primordium densities in 10-day-old seedlings of *iaa28-1* compared to Ws grown on control medium. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ by two-sided t test; $n > 10$.

(G) Time-lapse analysis from 0 to 25 hr after germination showing the percentage of seedlings ($n \geq 100$) with pDR5::GUS staining in the basal meristem in wild-type (blue squares) or *iaa28-1* mutant (red circles) background. The experiment was repeated three times with synchronized seedlings.

(H and I) Microarray data of longitudinal root sections (meristem [MS], rapid elongation zone [EZ], corresponding to the basal meristem zone], late elongation [deceleration] zone [DZ], mature root [MR], and the lateral root zone [LRZ]) showing relative expression of GATA23 (H) and IAA28 (I) in the different longitudinal sections of the root.

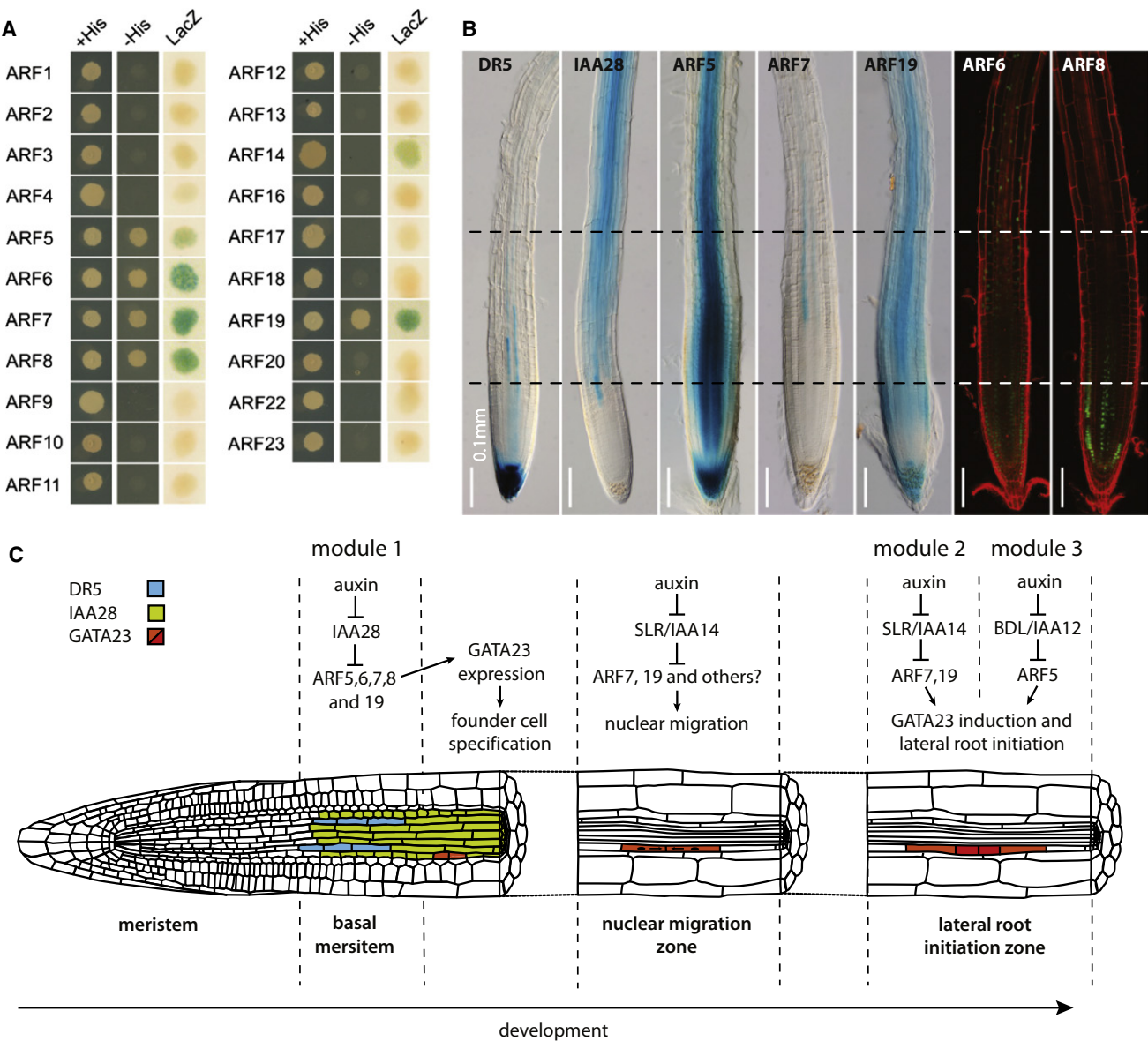


Figure 5. IAA28 Interacts with a Small Subset of ARF Proteins

(A) Yeast two-hybrid interaction study between IAA28 and all ARF proteins, showing interaction between IAA28 and ARF5, 6, 7, 8, and 19. (B) *pDR5::GUS*, *pIAA28::GUS*, *pARF5::GUS*, *pARF7::GUS*, *pARF19::GUS*, *pARF6::SV40-3×GFP*, and *pARF8::SV40-3×GFP* lines, with indication of the basal meristem region. Scale bars represent 0.1 mm. (C) Schematic representation of the auxin signaling modules acting successively during lateral root founder cell specification in the basal meristem (module 1) and nuclear migration and lateral root initiation processes (modules 2 and 3). As a consequence of module 1, GATA23 is hypothesized to be involved in the determination of founder cell identity in XPP cells.

root founder cell specification is further supported by the earlier reported *iaa28* mutant phenotype, which shows the most severe lateral root defect of all *aux/iaa* mutants next to *slr-1* [24]. Unlike most known auxin-inducible Aux/IAAs, both IAA28 expression levels and protein abundance have been reported to be reduced by auxin treatment in the root [12, 15, 24, 25]. Despite this differential response to auxin, it can be anticipated that every peak of auxin responsiveness in

the basal meristem will still result in removal of the Aux/IAA28 repressor, derepression of ARFs, and subsequent GATA23 activation. Although other auxin-controlled transcriptional networks might be involved during the founder cell specification process, the rescue of the IAA28 lateral rootless phenotype by transactivation of GATA23 in the pericycle cells strongly suggests that both players act in the same developmental pathway controlling lateral root formation.

(J) Lateral root primordium densities in 5-day-old seedlings of *iaa28-1* × J0121 × UAS::GATA23 compared to *iaa28-1* grown on control medium. **p* < 0.05, ***p* < 0.01, ****p* < 0.001 by two-sided *t* test; *n* > 10. Error bars represent means ± SEM.

The other known lateral rootless Aux/IAA mutant is *slr/iaa14*, but involvement of an SLR/IAA14-controlled auxin response module in founder cell specification can be excluded because of absence of SLR/IAA14 in the basal meristem and the demonstration that founder cell specification is still intact in *slr/iaa14* mutants using a dexamethasone-inducible mutant [11]. Here, we were able to specify more precisely the contribution of the SLR/IAA14-dependent signaling that appears to operate in the zone above the basal meristem and thus after appearance of *GATA23* expression and founder cell specification. Further research is needed, however, to determine whether *GATA23* is itself causally linked to the nuclear migration process or is responsible for an independent mechanism. At the least, our data show that IAA28-, SLR/IAA14-, and ARF7/ARF19-dependent signaling are required to guide the coordinated nuclear migration that precedes the actual asymmetric division. Earlier, pericycle founder cells were shown to undergo a PIN-dependent local accumulation of auxin prior to asymmetric cell division [22], which might result in the activation of different Aux/IAA-dependent modules (Figure 5C). During the same organogenetic process, auxin can thus signal through distinct modules depending on the phase of development. The identification of the *GATA23* transcription factor represents a first step toward the elucidation of the mechanisms that plants use to control lateral root spacing and guarantee an optimal distribution of new organs during the colonization of soil.

Experimental Procedures

Plant Growing Conditions and Time-Lapse Experiments

For all phenotypic analyses and time-lapse experiments, plants were grown on square plates (Greiner Labortechnik) with solid medium derived from standard MS medium incubated in a growth chamber under continuous light (110 $\mu\text{E}/\text{m}^2/\text{s}$ photosynthetically active radiation supplied by cool-white fluorescent tungsten tubes; Osram) at 22°C. For time-lapse experiments, plates with seeds were put in the growth room every 5 hr, and germination of seeds was synchronized by marking germinated seedlings for each marker line every 5 hr throughout the time-lapse. All seeds of each marker line that germinated at the same time were pooled and analyzed further.

Histochemical and Histological Analysis

GUS assays were performed as described previously [12]. For microscopic analysis, samples were cleared by mounting in 90% lactic acid (Acros Organics) or by clearing as described previously [21]. All samples were analyzed by differential interference contrast microscopy (Olympus BX51). For anatomical sections, GUS-stained samples were fixed overnight and embedded as described previously [26]. All pictures were processed with Adobe Photoshop 7.

In Vivo Root Confocal Imaging

For long-term confocal observations, seedlings were placed into a chambered borosilicate coverglass system (Nalge Nunc International) with a block of agar over the roots. This manner of visualization is advantageous in view of the prolonged observation. All supplements were added during the preparation of the agar blocks. The following transgenic lines were selected or generated from crosses and used in our observations: *slr1* [7] $\times$ 35S::LT16B-GFP $\times$ 35S::H2B-YFP [27]; *arf7arf19* [8] $\times$ 35S::LT16B-GFP $\times$ 35S::H2B-YFP; *iaa28* [24] $\times$ 35S::LT16B-GFP $\times$ 35S::H2B-YFP. Additionally, we generated a three-color imaging system based on GFP-labeled AtFH6 (35S promoter) for plasma membrane visualization, DR5-GFP-NLS, and RFP-labeled H2B (35S promoter). Temperature and light were kept as constant as possible during all observations.

Fluorescence imaging of roots was performed with an Axiovert 100M confocal laser scanning microscope with software package LSM 510 version 3.2 (Zeiss). For excitation of GFP, the 488 nm line of an argon laser was used; RFP excitation was produced by using a 520/20 nm excitation filter. Time-lapse series were typically collected at 4 min intervals and lasted

between 7 and 20 hr. Acquired images were processed and quantitatively analyzed with ImageJ v1.30 software.

Quantitative Real-Time PCR Confirmation

Quantitative real-time PCR analysis was performed as described previously [28]. RNA was extracted with an RNeasy kit (QIAGEN). Poly(dT) cDNA was prepared from 1 μg of total RNA with SuperScript III reverse transcriptase (Invitrogen) and analyzed on a LightCycler 480 apparatus (Roche Diagnostics) with the SYBR Green I Master kit (Roche Diagnostics) according to the manufacturer's instructions. Primer pairs were designed with Beacon Designer 4.0 (Premier Biosoft International) (Table S1). All individual reactions were performed in triplicate. Data were analyzed with qBase [29]. Expression levels were normalized to those of *EEF1 α 4* and *CDK1;1*.

Cloning and Construction of Expression Vectors

Donor vectors were created by cloning the promoter sequence for *GATA23* (SAP database [30]; <https://www.psb.ugent.be/promoter/>) or an upstream activating sequence (UAS) into the pDONRP4P1R donor vector and the genomic coding sequence into the pDONR221 donor vector. Expression clones were constructed by cloning the required fragments into pEX-K7SNFm14GW (promoter-NLS-GUS/GFP), pAGRIKOLA (RNAi lines), or pK7m34GW (translational GFP fusion and transactivation constructs) [31, 32]. The unique gene-specific tag sequence used to construct the RNAi line was TGCAATGCATGTGGAATTAGACACAGAAAACAGAGACGAT CAGAGTTATTGGGTATTCATATTATTCGACGCCACAAAAGCTTAGCCTCCA AGAAGATAAACCTATTATCATCATCACCGGTGGCGTGGCGGTGAAGAAA CGAAGGAGTCTAAAGGAGGAAGAACAAGCTGCTTTG (<http://www.catma.org/>). According to BLAST searches in *Arabidopsis*, this sequence only targets *GATA23*. For the *pARF6::SV40-3xGFP* and *pARF8::SV40-3xGFP* constructs, 2.1 kb upstream of each start codon was amplified using oligonucleotides pARF6-S/pARF6-AS or pARF8-S/pARF8-AS, subcloned, sequence verified, and cloned into a pGreenII-based vector containing the nuclear-localized 3 $\times$ GFP sequence and a NOS transcriptional terminator [33]. These constructs were introduced into *Agrobacterium* strain GV3101 harboring the helper plasmid pSoup. Two representative transgenic lines were analyzed for nuclear GFP expression in primary roots of homozygous T3 seedlings. All primers used are listed in Table S1.

Yeast Two-Hybrid Experiments

For Y2H analysis, full-length cDNA was used for IAA28 and ARF3, 13, 17, and 23 (ATG to stop). Partial cDNA encoding domains III and IV was used for the other ARFs and corresponded to the following region of the proteins: ARF1, 538-stop; ARF2, 727-stop; ARF4, 660-stop; ARF5, 788-stop; ARF6, 790-stop; ARF7, 1030-stop; ARF8, 699-stop; ARF9, 519-stop; ARF10, 575-stop; ARF11, 500-stop; ARF12, 500-stop; ARF14, 497-stop; ARF16, 570-stop; ARF18, 482-stop; ARF19, 948-stop; ARF20, 484-stop; ARF22, 498-stop. The cDNA or partial cDNA was cloned directionally in pENTR/D-Topo (Invitrogen) and then transferred by recombination to a Gateway-compatible pACT2-based vector for IAA28 (downstream of Gal4-AD) or a pGBKT7-based vector for the ARFs (downstream of Gal4-BD) [34]. For the actual Y2H screening, plasmids encoding the baits were transformed into the yeast strain PJ69-4 α (MAT α ; trp1-901, leu2-3,112, ura3-52, his3-200, gal4 Δ , gal80 Δ , LYS2::GAL1-HIS3, GAL2-ADE2, met2::GAL7-lacZ) and PJ69-4 α (MAT α ; trp1-901, leu2-3,112, ura3-52, his3-200, gal4 Δ , gal80 Δ , LYS2::GAL1-HIS3, GAL2-ADE2, met2::GAL7-lacZ) by the LiAc method. Interactions between proteins were assayed by the mating method. All bait yeast cultures were inoculated in 200 μl SD medium without Trp in a 96-well microtiter plate (Falcon), and one prey yeast culture was inoculated in 50 ml SD medium without Leu. To scale up the yeast cultures, 20 μl of each culture grown for 2 days at 30°C were added to a microtiter plate containing 125 μl of bacto yeast extract (10 g/l), bacto peptone (10 g/l), and dextrose (20%) (YPD) medium and again grown for 24 hr at 30°C. The YPD medium was replaced by SD medium without Leu and Trp. Diploid strains grown in a 96-well microtiter plate (Nunc) for 2 days at 30°C were diluted to OD₆₀₀ = 0.2 and then added to a 96-well microtiter plate (Falcon) containing either 190 μl SD medium without Leu and Trp but with His (as control) or SD medium without Leu, Trp, and His. OD was measured after 2 days of incubation at 30°C. Interactions were scored as positive when the ratio of the OD₆₀₀ of SD medium without Leu, Trp, and His on that of the SD medium without Leu and Trp exceeded 70%. For the LacZ test, 3 μl of the OD₆₀₀ = 0.2 dilution was spotted on a Hybond-N+ membrane (GE Healthcare) on a YPD agar plate. After incubation for 2 days at 30°C, the membrane was snap frozen in liquid nitrogen and placed on two Whatman papers soaked in 6 ml of Z buffer (16.1 g Na₂HPO₄ 7H₂O, 5.5 g NaH₂PO₄ H₂O,

0.750 g KCl, and 0.246 g MgSO₄ 7H₂O) with 11 μ l β -mercaptoethanol and 100 μ l 4% 5-bromo-4-chloro-3-indolyl- β -D-galactopyranoside. After incubation for 6 hr at 37°C, colonies were scored visually for blue staining.

Accession Numbers

Microarray data reported herein have been deposited at ArrayExpress (<http://www.ebi.ac.uk/arrayexpress/>) with the accession number E-MEXP-2912.

Supplemental Information

Supplemental Information includes five figures, three tables, Supplemental Experimental Procedures, and three movies and can be found with this article online at doi:10.1016/j.cub.2010.09.007.

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